PH.45320 **FALL 2026**

Static Cosmology

Lecture 4: The 3.5 K Cosmic Medium

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Overview

Kirchhoff equilibrium connects emissivity and absorption to a 3.5 K Planck spectrum.

1 Observational setup

Source coordinates, observing band, selection function, and calibration state are fixed at the outset.

$$B_\nu(T_m) = 2h\nu^3/c^2(e^{h\nu/k_BT_m} - 1)^{-1}$$

2 Transport or equilibrium equation

The governing equation is solved first in a homogeneous limit and then with the stated spatial dependence.

$$T_m = 3.5K$$

3 Connection to data

Each model quantity is tied to a measured flux, spectrum, angle, count, or temperature.

$$j_\nu = \alpha_\nu B_\nu(T_m)$$

4 Degeneracies

Calibration, population, and medium parameters are varied separately to expose their degeneracies.

5 Example

The worked example carries units and uncertainty from the input catalog to the reported result.

Checks

- Verify dimensions and sign conventions.
- State the boundary conditions used in each limit.
- Report numerical precision separately from model uncertainty.

Reading

- Jeong, ch. 4
- CMT-9 T block
- Thermal Gain r12 note